Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/30/2026
PI Name:	Yeung, Catherine Ph.D.		
Project Title:	Extended Culture of Kidney MPS and Organoids to Model Acute and Chronic Exposure to Drugs and Environmental Toxins		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	11/15/2022	**End Date:**	11/14/2026
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
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Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:	No changes in key personnel		
COI Name (Institution):	Bammler, Theodor Ph.D. (University of Washington) Freedman, Benjamin Ph.D. (University of Washington) Himmelfarb, Jonathan Ph.D. (University of Washington) Kelly, Edward Ph.D. (University of Washington) Thummel, Kenneth Ph.D. (University of Washington) Zelnick, Leila Ph.D. (University of Washington)		
Grant/Contract No.:	80ARC023CA001		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Kidney disease is a public health problem affecting 850 million people worldwide, 37 million people in the US adult population, and is the 9th leading cause of death. Loss of kidney function is often associated with chronic hypertension, diabetes, or exposure to pathogens, drugs, or environmental chemicals, and in many cases, takes many months or years to manifest clinically. Finding efficacious therapeutics and preventing disease progression require fundamentally new strategies, assays, and models of the kidney to improve patient outcomes. Until recently, the lack of human in vitro models that recapitulate critical aspects of kidney physiology, mimic the unique complexities of specific nephron segments, model disease heterogeneity, or assess reparative mechanisms in response to injury has hindered progress. Our team has approached the scientific challenges of studying kidney disease pathophysiology, novel therapeutic agents, and drug/ environmental toxicity by establishing two innovative human models: the proximal tubule kidney chip microphysiological system (PTEC MPS) and the iPSC (induced pluripotent stem cell)-derived kidney organoid. We have developed validation assays for short-duration (2-10 day) exposures in both ground-based laboratories and aboard the International Space Station. The goal of this project is to demonstrate models with extended (6 month) longevity to better understand the role of chronic stressor exposure on kidney disease initiation and progression. Based on previous studies, we hypothesize that human kidney systems will remain viable and functionally stable for at least 6 months and will respond physiologically to extended exposure to stressors. Phase I – Description and Tasks. Establish extended longevity and functionality (6 months) of kidney microphysiological systems and organoids. The contractor shall: • Determine longevity of the proximal tubule kidney chip (PTEC) MPS at 2.5, 5, and 6 months by employing cell viability (live/dead staining), metabolic function (conversion of 25(OH)VitaminD3 to 1a25(OH)2 VitaminD3), toxic response (resulting in KIM-1 [kidney injury molecule-1] expression), stable expression of immunomodulators (e.g., IL[interleukin]-6, IL-23A, TNF [tumor necrosis factor], IL-32), and global RNA transcript profile. • Determine longevity and stable function of kidney organoids at 2.5, 5, and 6 months by conducting observation of distinct tubules with specific multiple nephron and stromal lineages in segmented structures (immunocytochemical staining), and stable global single cell RNA transcript profile. Phase II – Description and Tasks. Test structural and functional outcomes after sustained exposure to drugs, environmental toxins, and pathogens and post-exposure cellular recovery. The contractor shall: • Determine functional outcomes of PTEC MPS after exposure to agents known to cause nephrotoxicity after 5 months exposure (PMB [polymyxin B] or other nephrotoxin) followed by 1 month recovery by evaluating structural derangement (de-polarization of transporter proteins), production of injury biomarkers (KIM-1, HO-1 [heme oxygenase-1], cystatin), and altered regulation of immunomodulators (e.g., IL-6, IL-23A, TNF, IL-32). • Evaluate kidney organoid responses to pathogenic stimuli and known nephrotoxins using fluorescent reporters for real-time detection of infection and injury in response to acute (3 days) and extended (5 months) exposure followed by 1 month of recovery.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Developing a kidney microphysiological system and kidney organoids with extended longevity will allow us to model the effects of stressors like drugs, environmental toxins, pathogens, and microgravity on kidney cell structure and function, and to determine if, and how, kidney cells can recover from stress. A better understanding of kidney injury and recovery from chronic stressor exposure can accelerate the development of better and safer medications and treatments to prevent and cure kidney diseases.
Task Progress:	Previously, we were able to maintain viable human kidney organoids for six months and developed an associated transcriptomic atlas of aging using this system, with corresponding immunofluorescence images (Phase I). These data were presented at national meetings, and we are working to publish a manuscript. Our ongoing experiments are focused on understanding how nephrotoxicity and pathogens affect kidney organoids in both acute and chronic settings (Phase II). We have now completed a 180-day experiment with 5-month continuous treatment of either a chemical nephrotoxicant or an inflammatory cytokine, each at three different concentrations, followed by a recovery period. By imaging every few days, we have obtained detailed time-lapse images, which correspond to immunofluorescence images at endpoint. Dose-dependent toxicities are being observed, which we are analyzing to determine sensitivity to concentration and the total time of exposure. We are in the process of repeating the experiment. We have validated antibodies for detection of kidney injury biomarkers, which we intend to test in this system. Earlier experiments demonstrated that kidney chips could be successfully cultured with >90% viability for more than 365 days and demonstrate the capacity for healing gaps in the monolayer. Functional tests (KIM-1 response to polymyxin B exposure and Vitamin D metabolism) and transcriptomics conducted at 6 months showed mixed results. IL-6 secretion and epithelial cell identity were stably maintained for 6 months; however, response to polymyxin B and Vitamin D metabolism were suppressed (but still present). This may be due to de-induction of response pathways after extended culture without physiological stimulus, or due to aging of the cells in culture. These studies are currently being repeated with experimental modifications – including extended, lower-concentration exposure to polymyxin B and CYP24A1 induction with 1,25(OH)2 Vitamin D3 – to increase metabolism of the 25-OH Vitamin D3 substrate.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Other Journals or Periodicals	Spennati G, Regier MC, Ward HH, Das V, Karihaloo A, Freedman BS. "Elevated glucose in organoids induces a tissue-intrinsic inflammatory response driving epithelial detachment." bioRxiv preprint server. Posted September 8, 2025. https://doi.org/10.1101/2025.09.05.669416 , Sep-2025
Articles in Peer-reviewed Journals	Chalker B, Tsang YP, Mahadeo A, Yeung CK, Himmelfarb J, Kelly EJ. "Isolation of primary human proximal tubule epithelial cells and their use in creating a microphysiological model of the renal proximal tubule." J Vis Exp. 2025 May 9:(219). http://dx.doi.org/10.3791/67997 ; PMID: 40418667 , May-2025
Articles in Peer-reviewed Journals	Tsang YP, Aryeh KS, Wang K, Himmelfarb J, Yeung CK, Kelly EJ. "Enhancing therapeutic strategies and drug development for patients with kidney disease." Expert Opin Drug Saf. 2025 Jul 4;1-26. Review. https://doi.org/10.1080/14740338.2025.2525970 ; PMID: 40568828; PMCID: PMC12285105 , Jul-2025